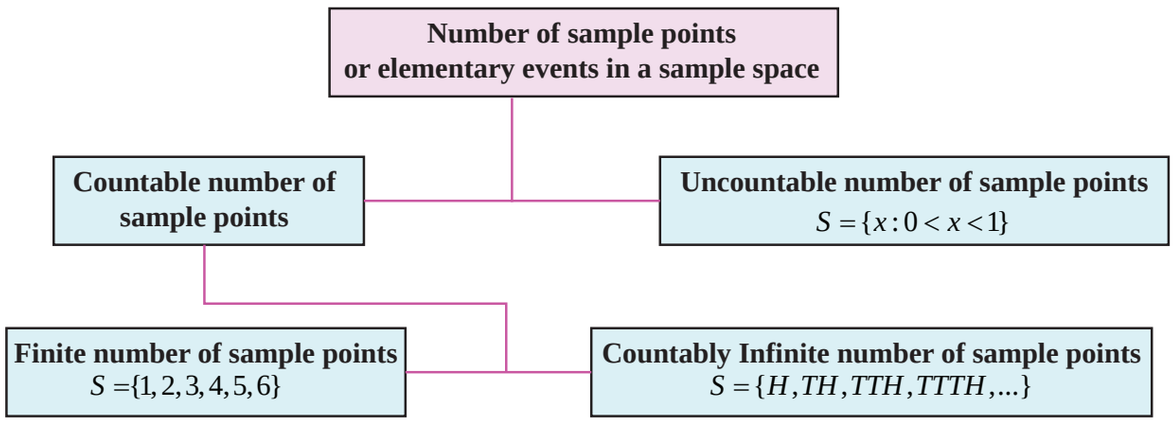
From (2) and (3), one need to distinguish between two types of infinite sets, where one type is significantly 'larger' than the other. In particular, S in (2) is called **countably infinite**, while the S in (3) is called **uncountably infinite**. The fact that one can list the elements of a countably infinite set means that the set can be put in one-to-one correspondence with natural numbers $\mathbb{N}$. On the other hand, you cannot list the elements in uncountable set.

From the above example, one can understand that the sample space may consist of countable or uncountable number of elementary events.



12.3 Finite sample space

In this section we restrict our sample spaces that have at most a finite number of points.

Types of events

Let us now define some of the important types of events, which are used frequently in this chapter.

- Sure event or certain event
- Complementary event
- Mutually inclusive event
- Equally likely events
- Impossible event
- Mutually exclusive events
- Exhaustive events
- Independent events (defined after learning the concepts of probability)

Definition 12.6

When the sample space is finite, any subset of the sample space is an **event**. That is, all elements of the power set $\mathcal{P}(S)$ of the sample space are defined as **events**. An event is a collection of sample points or elementary events.

The sample space S is called **sure event or certain event.** The null set $\varnothing$ in S is called an **impossible event.**

Definition 12.7

For every event A, there corresponds another event $\overline{A}$ is called the **complementary event** to A. It is also called the event 'not A'.

Illustration 12.2

Suppose a sample space S is given by $S = \{1,2,3,4\}$.

Let the set of all possible subsets of S (the power set of S) be $\mathcal{P}(S)$.

$$\mathcal{P}(S) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\},$$
$$\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \{1, 2, 3, 4\}\}$$

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